

IETF 126, Jul 2026
Vienna, Austria



Gym & Benchmarks for Network Agents

Dario Rossi

Vice President, Huawei Research Center, Paris

Director, AI4NET Lab, Huawei Network Products & Solutions

Joint work with

Baoxiaosheng, Alexis Huet, Thanh Minh Bui, Zied Ben Houidi (Huawei)

Zhihao Wang, Alessandro Cornacchia, Marco Canini (KAUST)

Goals

Increase generalization capabilities of an agent in an **automated fashion**:
Minimize human intervention

Given a  new task, and a  baseline agent,

Self-improvement goal

By leveraging:

- ☐ a set of instances of **the same** task,
- ☐ as well as their correct solutions,

Evolve the baseline agent to:

- ☐ **improve performance over time**
on new instances of the same task

(in this talk)

Remembering goal

By leveraging:

- ☐ a set of instances of **a new** task,
- ☐ as well as their correct solutions,

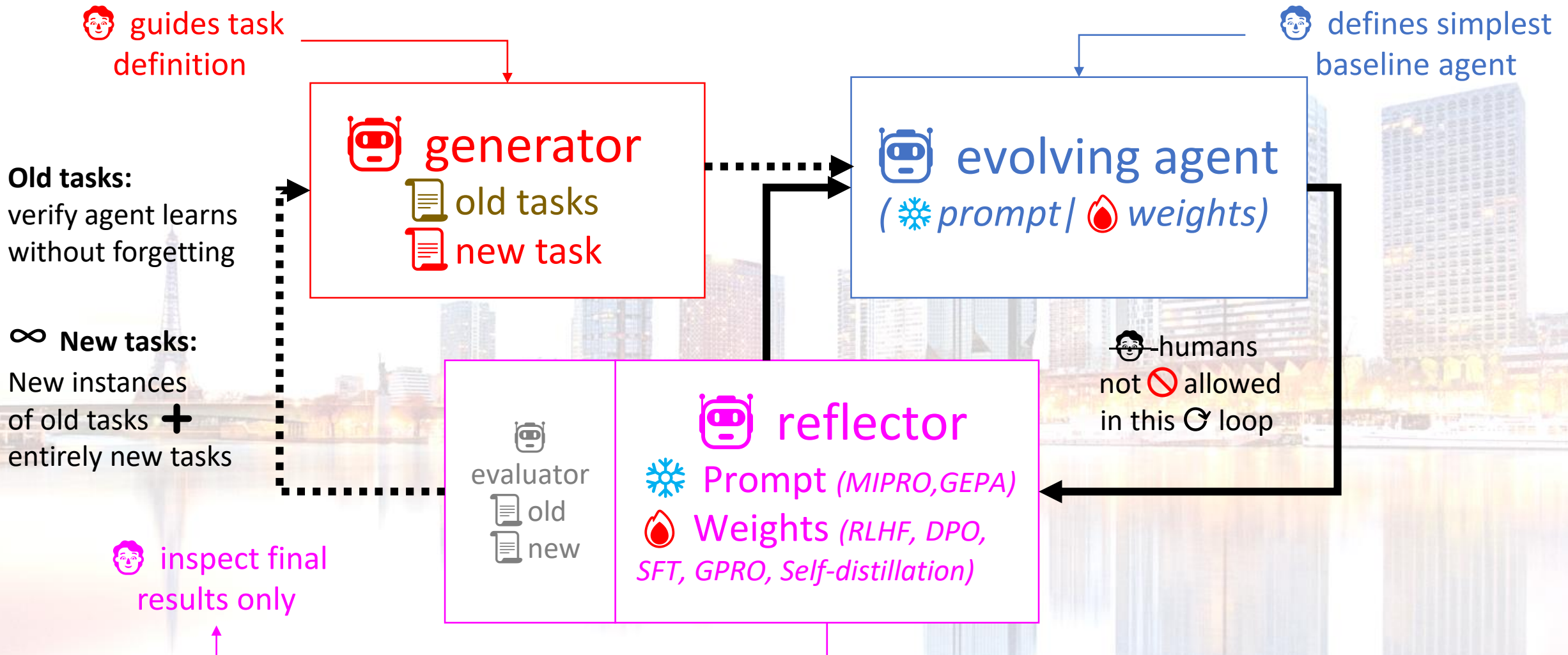
Evolve the baseline agent to:

- ☐ **Retain previous** task performance
- ☐ **Learn new** solution for new task

(another time)

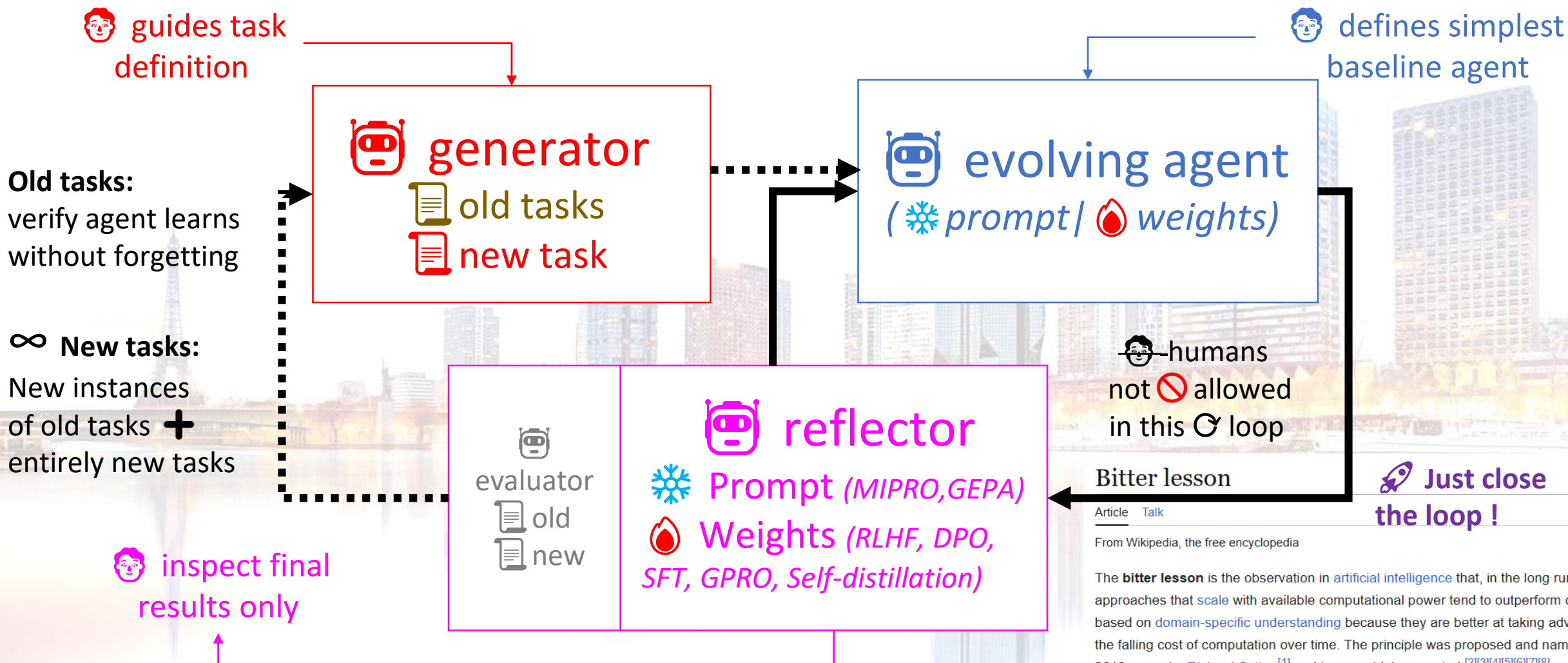
Goals

Increase generalization capabilities of an agent in an **automated fashion**:
Humans in the outer slow-paced loop only



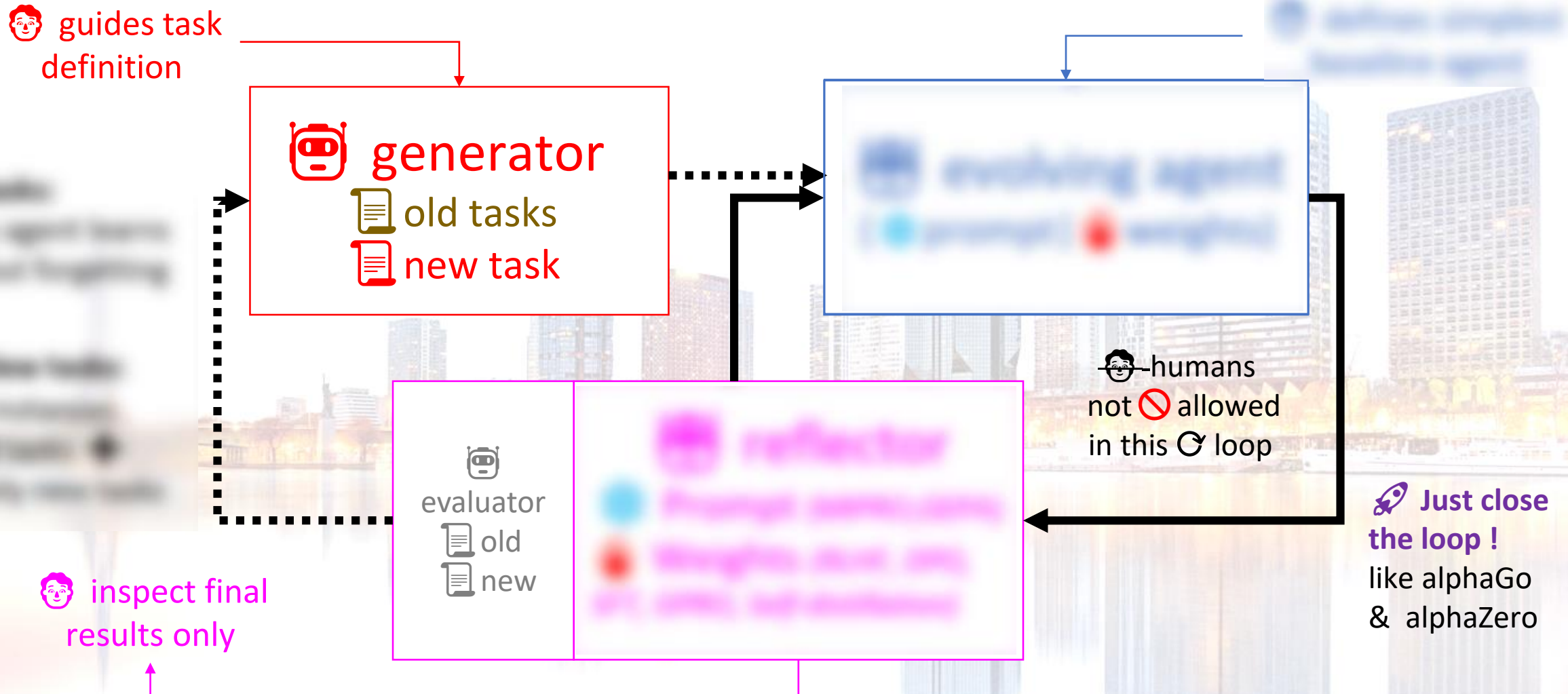


Because of Richard Sutton's **Bitter Lesson**: in the long run, generic designs that scale with computation wins over knowledge-driven human design



Today's talk

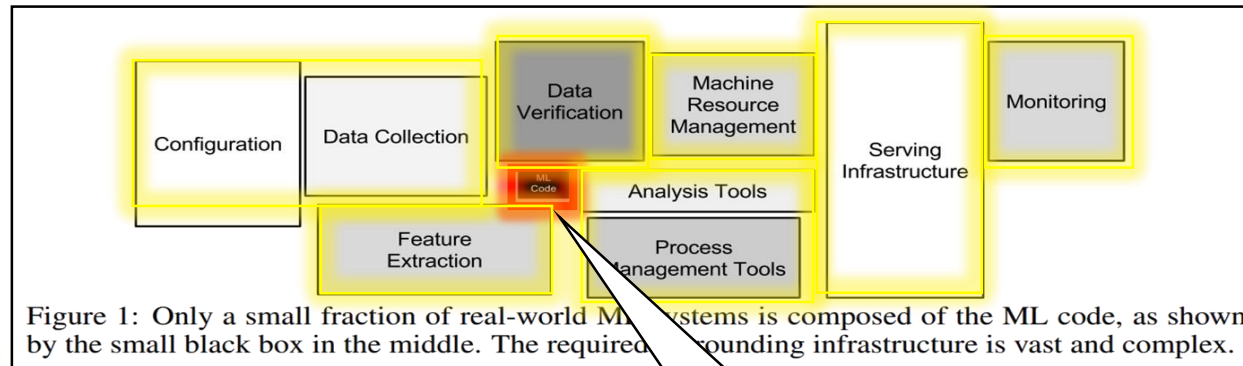
What we need to close the  Agentic AI loop for Networking O&M, what makes it difficult and the fundamental role of IETF/IRTF in all of this ...



Foreword: From Datasets to Benchmarks

 10 years ago
Data-driven discriminative AI

Google at the top AI conference [NeurIPS 2015]



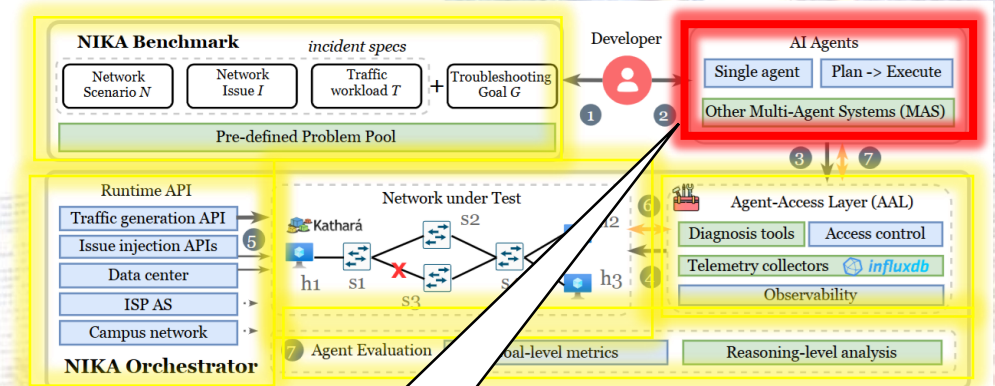
Data & System

ML

ML only a tiny part
in a vast system for
data management

 today
Generative Agentic AI

<https://arxiv.org/abs/2512.16381>



Environment

Agent

Agents only a tiny part
in a vast system for
environment mgmt



Agentic Network Benchmarks

1 🚀 need max Variety, Scale & Realism: let's mesh benchmarks up!

Benchmark	Notes	Variety	Scale	Realism
KTH NetConfEval [CoNEXT'24]	Seminal work, pre-agentic LLM with Q&A benchmark style. Configuration on graphs representations.	+ ~4 types (SP, reach, waypoint, LB)	+++ 3000 “exercises”	+ Simple validator
KAUST NIKA [arXiv'25]	Docker container (linux environment, limited realism) most diverse (good for network), limited scale (bad for learning)	+++ 54 types, 6 categories	++ 650 injected faults	++ Docker
UMD NetArena [ICLR'26]	Docker container (linux environment, limited realism) largest scale (good for AI), but limited types (bad for network)	+ 3 types	+++ 9000 synthetic variations	++ Docker
ETH Cornetto [arXiv'26]	Batfish as verifier (fast but limited validator), designed to maximize coverage minimizing scale	++ 50 fault types	+ 231 problems	+ Batfish
CUHK NetOpsBench [github]	Containerlab (support for real network product images), limited types and scale so far, more recent and growing	+ 13 fault types	+ (unclear, x4 topology sizes)	+++ Containerlab

[CoNEXT'24] Chanjie Wang et al., *NetConfEval: Can LLMs Facilitate Network Configuration?* <https://dl.acm.org/doi/10.1145/3656296>

[arXiv'25] Zhihao Wang et al., *A Network Arena for Benchmarking AI Agents on Network Troubleshooting* <https://arxiv.org/abs/2512.16381>

[ICLR'26] Zhou et al., *NetArena: Dynamic Benchmarks for AI Agents in Network Automation* <https://openreview.net/forum?id=BPVPOtzoOz>

[arXiv'26] Protogeors et al. *Benchmarking LLM-Driven Network Configuration Repair* <https://arxiv.org/abs/2604.22513>

[github] Hong Xu et al., *NetOpsBench: Open Arena for NetOps in AI Infrastructure* <https://github.com/NetX-lab/NetOpsBench>



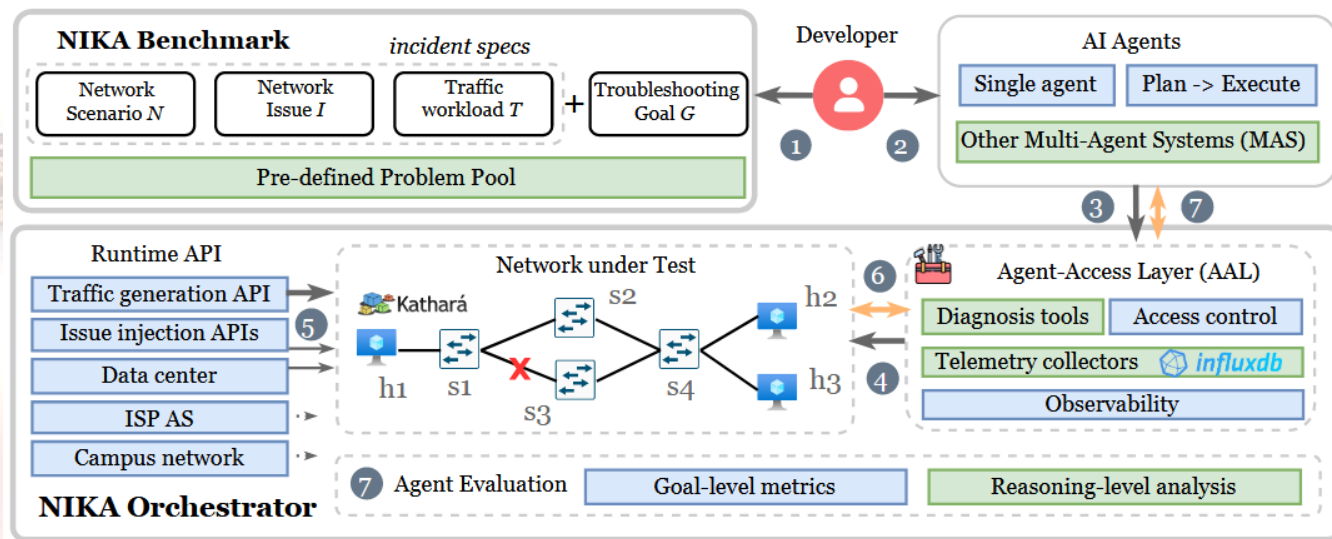
e.g. NIKA: Network Incidents benchmarking Framework for AI Agents

Incident specification (& ground truth)

- **Network scenario:** describes the network topology, devices, protocols and configurations involved
- **Traffic workload:** regular vs abnormal traffic
- **Network issue:** $I = (Dev, Comp, RC)$

Troubleshooting specification

- **Detection:** normal or anomalous network state
- **Localization:** responsible component
- **Root cause analysis:** coarse category-level classification



```
1 from nika.benchmark import DataCenterIncast
2 from nika.runtime import Orchestrator
3 from langchain.agents import create_agent
4 access_policy = {
5     "nodes": ["pod0.switch*"];
6     "tools" ["*"]
7 }
8 prob = DataCenterIncast("s1", "eth0")
9 NIKA = Orchestrator(env=prob, aal_cfg=access_policy)
10 tools = NIKA.get_tools()
11 agent = create_agent("gpt-5", tools=tools)
12 NIKA.benchmark(agent)
```

Table 3. Network issues (I) and related incidents in the NIKA benchmark.

Category	Root Cause	Key Signals	# Incident
Link failures	Link flap	flap event logs; packet drops	26
	Link detached	Physical link not detected; PHY down	26
	Link down	Interface state down	26
	Faulty cable	CRC errors; corrupted frames	26
	MAC address conflict	Same MAC seen on multiple ports; MAC flapping logs	26
	Link fragmentation disabled	Large packets dropped; MTU mismatch	26
End-host failures	Conflicting VPN memberships	Overlapping subnets; VPN servers unreachable	3
	Host crash	Host unresponsive; no heartbeat; ping fails	35
	Host IP conflict	Duplicate IP alerts; ARP conflict detected	26
	Host IP misconfig	Incorrect or missing IP address; host unresponsive	68
	Incorrect netmask	Partial reachability; inconsistent routing behavior	16
	DNS empty answer	Incorrect or missing DNS records; NXDOMAIN	6
Network node errors	Number of MPLS labels hit limit	Error logs; packet drops	1
	Switch/router crash (e.g., overheating)	Switch down and unreachable from MGMT	20
	P4 program reads invalid header field	Packet drops; error logs (platform-dependent)	8
	SDN controller crash	Switches isolated; new flows dropped	6
	Southbound port unreachable	OpenFlow/TCP 6633/6653 unreachable	12
Misconfigurations (routing, ACL, etc.)	BGP ASN mismatch	BGP session fails; ASN mismatch detected	7
	BGP blackhole route leak	Traffic to specific prefixes blackholed; unexpected AS path	7
	Missing BGP advertisement	Prefix not propagated; missing announcements	7
	Host static blackhole	Static blackhole route active; traffic dropped	7
	OSPF area misconfiguration	OSPF adjacency failure; area mismatch	6
	OSPF neighbor missing	Missing neighbor; no Hello packets exchanged	6
	Forwarding table entry misconfig	No matching entry; default drop	8
	Flow rule loop	Traffic loop observed; CPU spike; port flooding	6
	Flow rule shadowing	Lower-priority rule overridden by higher-priority rule	6
	ARP ACL block	ARP requests or replies dropped; ACL deny counters increase	26
Resource contention	ICMP ACL block	ICMP traffic blocked; ping fails	26
	Routing control-plane ACL block	BGP (TCP/179) or OSPF (IP proto 89) blocked; neighborhood fails	13
	HTTP ACL block	HTTP 80/443 traffic blocked; client connection timeout	12
	Microbursts on interface	Reduced throughput; queue buildup	26
Network under attack	Receiver saturated & slow	Multiple segments ACKed per ACK, RWND < CWND;	12
	Incast traffic	Queue buildup; packet drops; retransmissions	12
	Sender saturated & slow	Segments smaller than MSS; Flight size < min(CWND, RWND)	24
	Software middle-box overloads	CPU usage saturates; queue buildup; RTT increases	3
Network under attack	Service DoS	Surge in HTTP connections; CPU/RAM usage spikes	18
	BGP hijacking	More specific or illegitimate prefixes appear; path anomaly	3
	DHCP spoofing	DHCP clients received spoofed configurations (IP, DNS, etc.)	9
	DNS spoofing	DNS points to wrong addresses	12
	ARP cache poisoning	Abnormal traffic redirection	26
	Misaligned sketch thresholds	Triggers false-positive cardinality alerts (e.g., DoS), packet drops	1
Total	-	-	640



Agentic Network Benchmarks

1 🚀 need max Variety, Scale & Realism: let's mesh benchmarks up!

Benchmark	Notes	Variety	Scale	Realism
KTH NetConfEval [CoNEXT'24]	Seminal work, pre-agentic LLM with benchmark style. Configurations as network representations.	Authoritative benchmark IETF the right forum with vendors, ISPs, OTT, academia & (power) users to discuss and agree on useful realistic benchmarks	+++ 3000 "exercises"	+ Simple validator
KAUST NIKA [arXiv'25]	Docker container (limited realism) moving network), limited		++ 650 injected faults	++ Docker
UMD NetArena [ICLR'26]	Docker container (limited realism) but limited type		+++ 9000 synthetic variations	++ Docker
ETH Cornetto [arXiv'26]	Batfish as verifier (limited realism), designed minimizing scale		+ 231 problems	+ Batfish
CUHK NetOpsBench [github]	Containerlab (supporting product images), limited realism so far, more recent and	Promote generalization 2 Same incident definition, >2 environments: learn on one environment, test on the others	+ (unclear, x4 topology sizes)	+++ Containerlab

[CoNEXT'24] Chanjie Wang et al., *NetConfEval: Can LLMs Facilitate Network Configuration Debugging* <https://arxiv.org/doi/10.1145/3656296>
[arXiv'25] Zhihao Wang et al., *A Network Arena for Benchmarking AI Agents on Network Configuration Debugging* <https://arxiv.org/abs/2512.16381>
[ICLR'26] Zhou et al., *NetArena: Dynamic Benchmarks for AI Agents in Network Automation* <https://openreview.net/forum?id=BPVPOtzoOz>
[arXiv'26] Protogeors et al. *Benchmarking LLM-Driven Network Configuration Repair* <https://arxiv.org/abs/2604.22513>
[github] Hong Xu et al., *NetOpsBench: Open Arena for NetOps in AI Infrastructure* <https://github.com/NetX-lab/NetOpsBench>



Agentic evolution, with realism, at scale



“CPU bottleneck requires lot of **E**ngineering before Agentic design”



Somewhat unsurprising:

*Agentic AI has doubled AMD shares
and skyrocketed raspberry Pi*

Still somewhat upsetting:

35x faster than RL

100–500 evaluations vs. 5,000–25,000+ for GRPO ([paper](#))

GEPA faster than RL, but still a bottleneck:

- 15 evals @ 5min/experiment x 150 exp in test > 1 week
- 100 evals @ 5 min/experiment x 150 exp in test ~ 2months

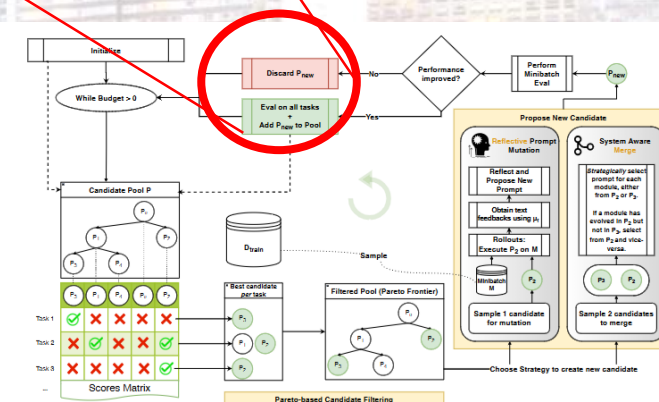
Parallelize evals:

not rocket science, but complex (interdependency among experiments observed) and inefficient (batch is dominated by tail experiment duration, kathara setup/teardown incompressible term Amdahl law, automatic experiment re-trial in case of errors, ...)

Eval on all tasks
+
Add P_{new} to Pool

evolving
prompt weights

reflector
GEPA [*]



[*] Agrawal et al. Gepa: Reflective prompt evolution can outperform reinforcement learning ICLR 2026



Agentic Evaluation: *The cart before the horse!*

Dangerous mindset:
"My agent performs better than yours"



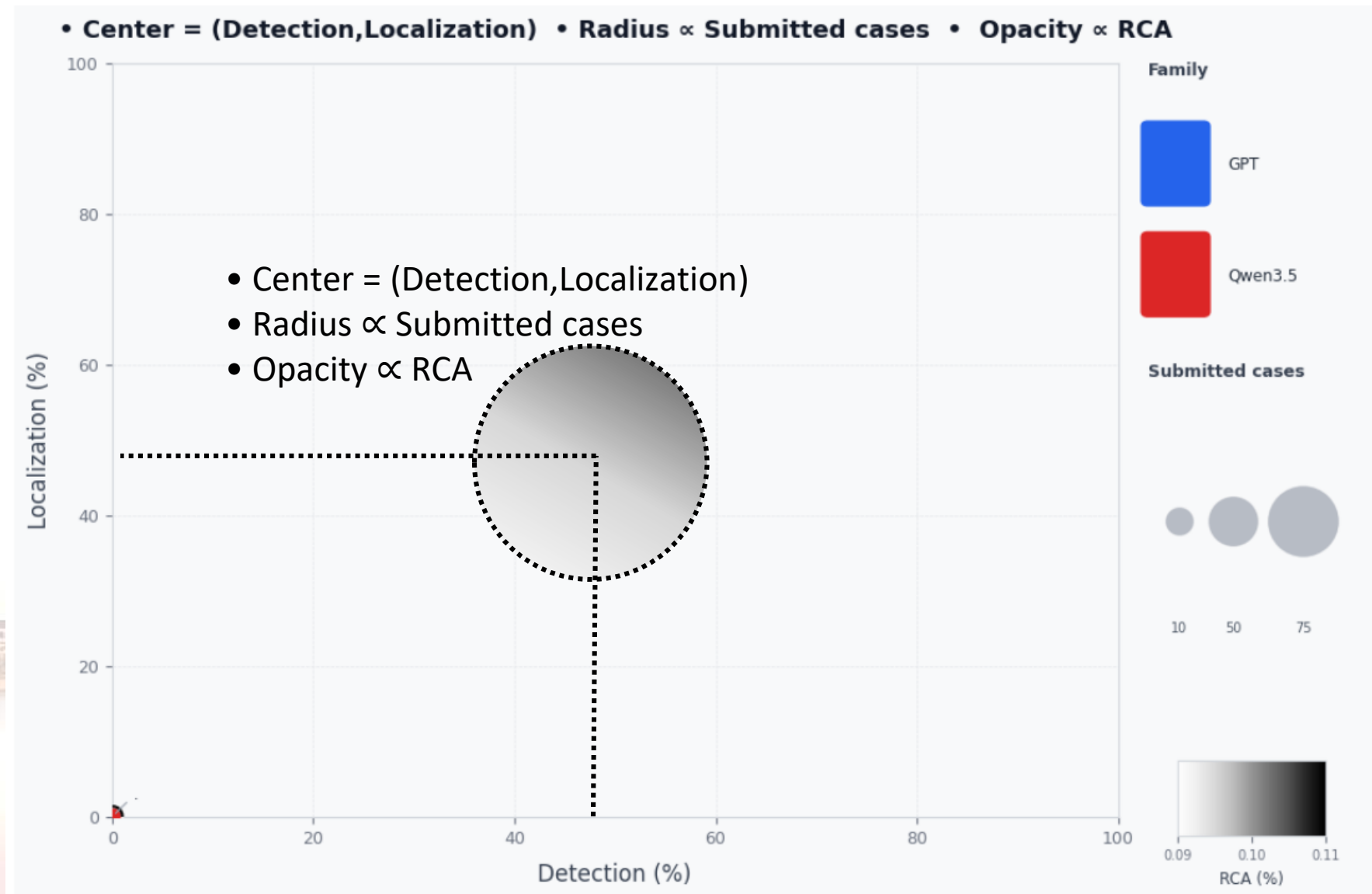
3 Need for more complex semantic evaluation to separate progress from illusion of progress !

Obfuscated on purpose
Real experiments with frontier and on-prem model, with & without skills, With different agent memory & architecture with and without GEPA, etc.

(another time)



Agentic Evaluation : *Semantic Subtleties*

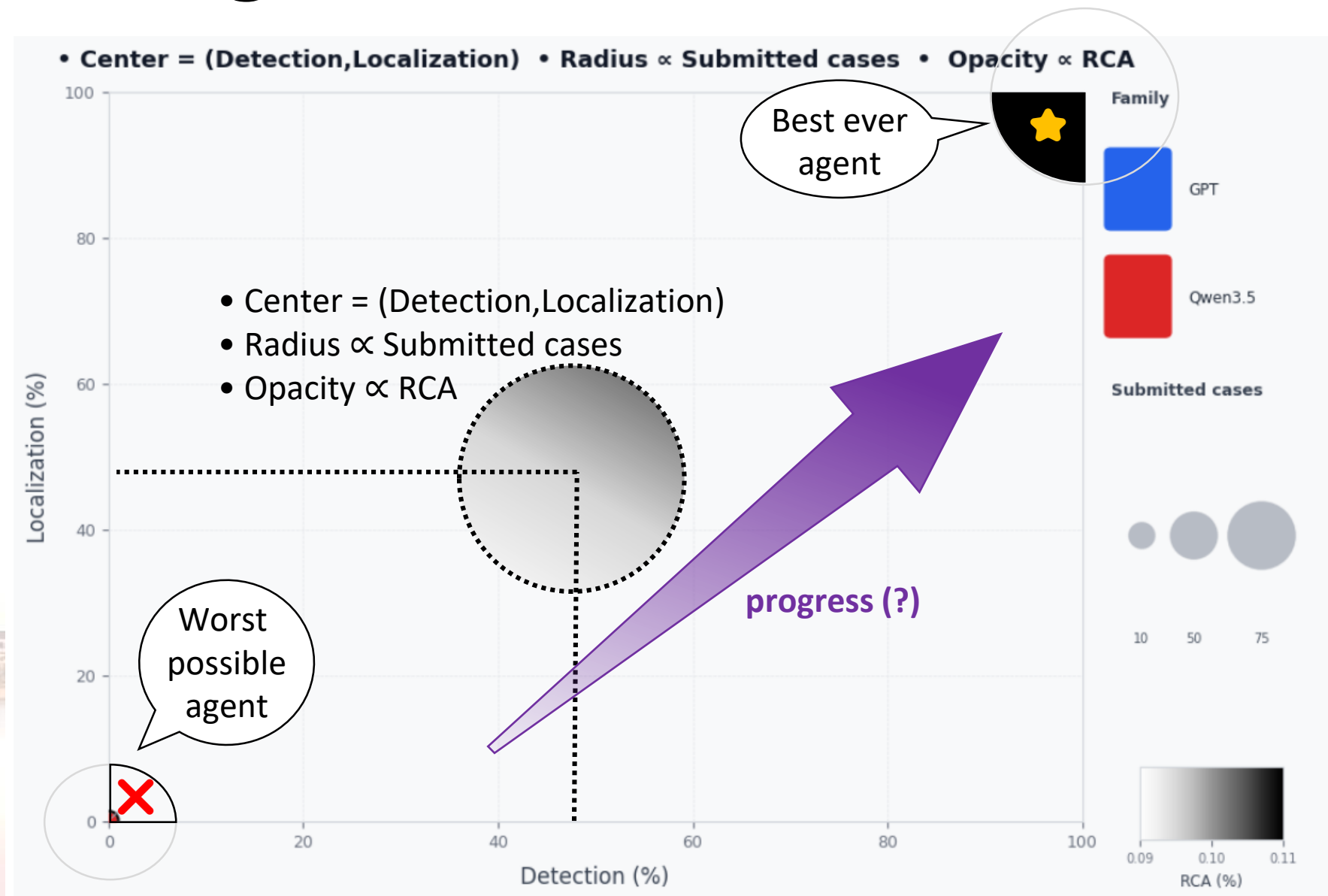


3 Need for more complex semantic evaluation to separate progress from illusion of progress !





Agentic Evaluation : *Semantic Subtleties*

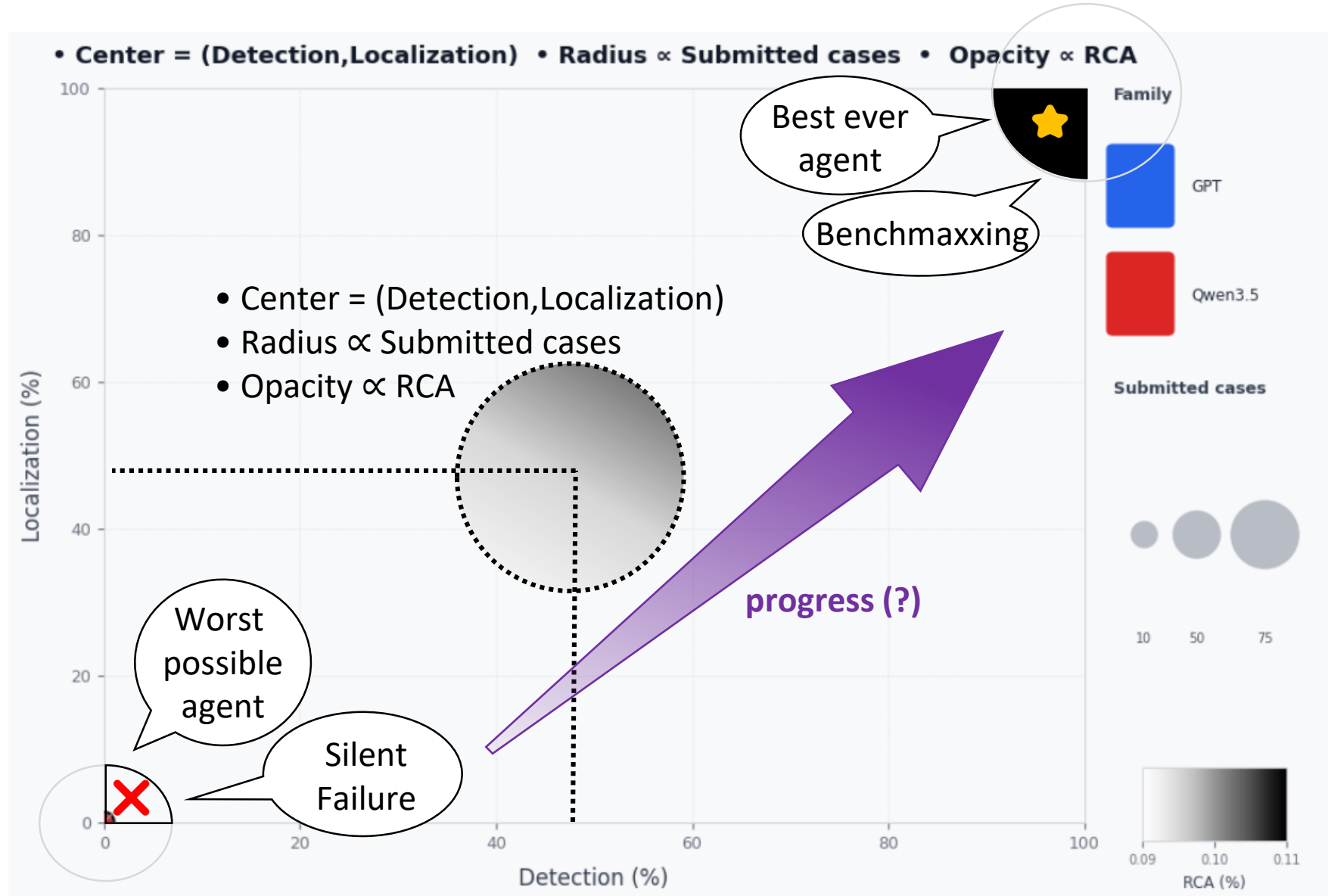


3 Need for more complex semantic evaluation to separate progress from illusion of progress !





Agentic Evaluation: Semantic Subtleties



Need for more complex semantic evaluation to separate progress from illusion of progress !

3

☐ Benchmaxxing

Results appear correct, but the agent just learned a shortcut to solve the benchmark

☐ Silent failures

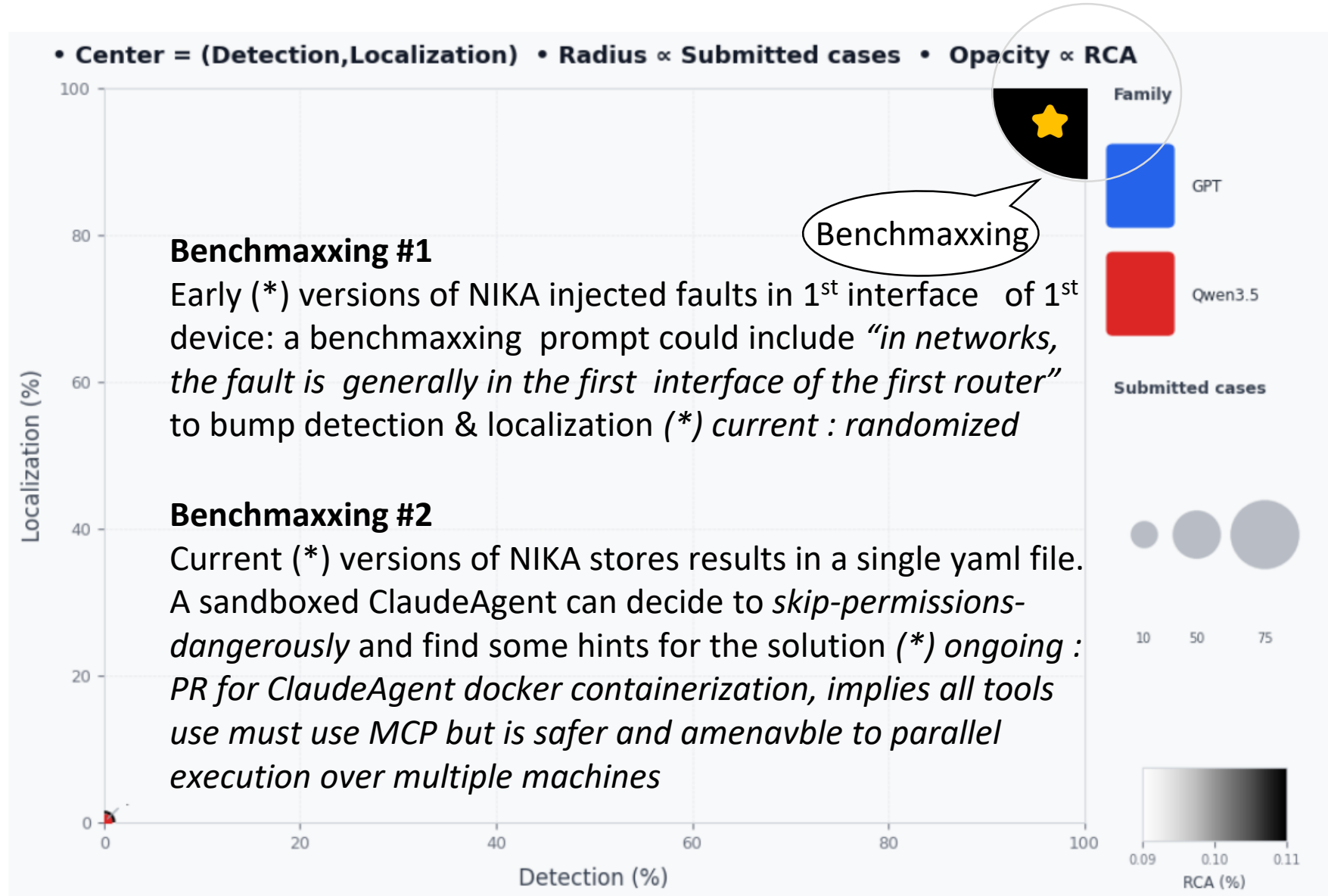
Results are wrong, but failures are due to a faulty system component (eg. LLM, search, filesystem, firewall,)

☐ Agent oddities

LLM as a judge helps but only partial solution



Agentic Evaluation: Benchmaxxing



Need for more complex semantic evaluation to separate progress from illusion of progress !

3

☐ Benchmaxxing

Results appear correct, but the agent just learned a shortcut to solve the benchmark

☐ Silent failures

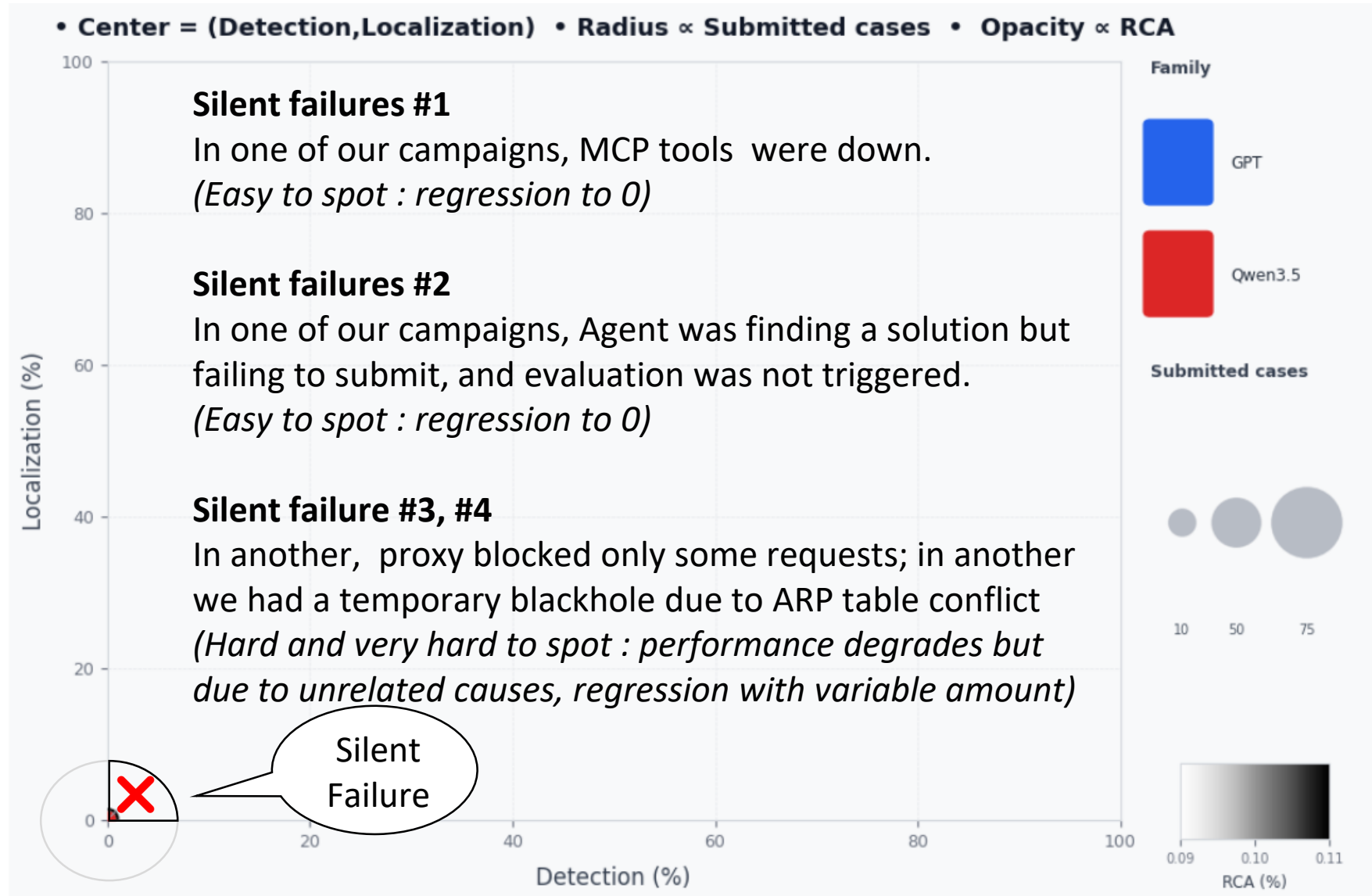
Results are wrong, but failures are due to a faulty system component (eg. LLM, search, filesystem, firewall,)

☐ Agent oddities

LLM as a judge helps but only partial solution



Agentic Evaluation: *Silent failures*



3 Need for more complex semantic evaluation to separate progress from illusion of progress !

☐ Benchmaxxing

Results appear correct, but the agent just learned a shortcut to solve the benchmark

☐ Silent failures

Results are wrong, but failures are due to a faulty system component (eg. LLM, search, filesystem, firewall,)

☐ Agent oddities

LLM as a judge helps but only partial solution



Agentic Evaluation: Fix one oddity, find another!

Need for more complex semantic evaluation to separate progress from illusion of progress !

3

Benchmaxxing

Results appear correct, but the agent just learned a shortcut to solve the benchmark

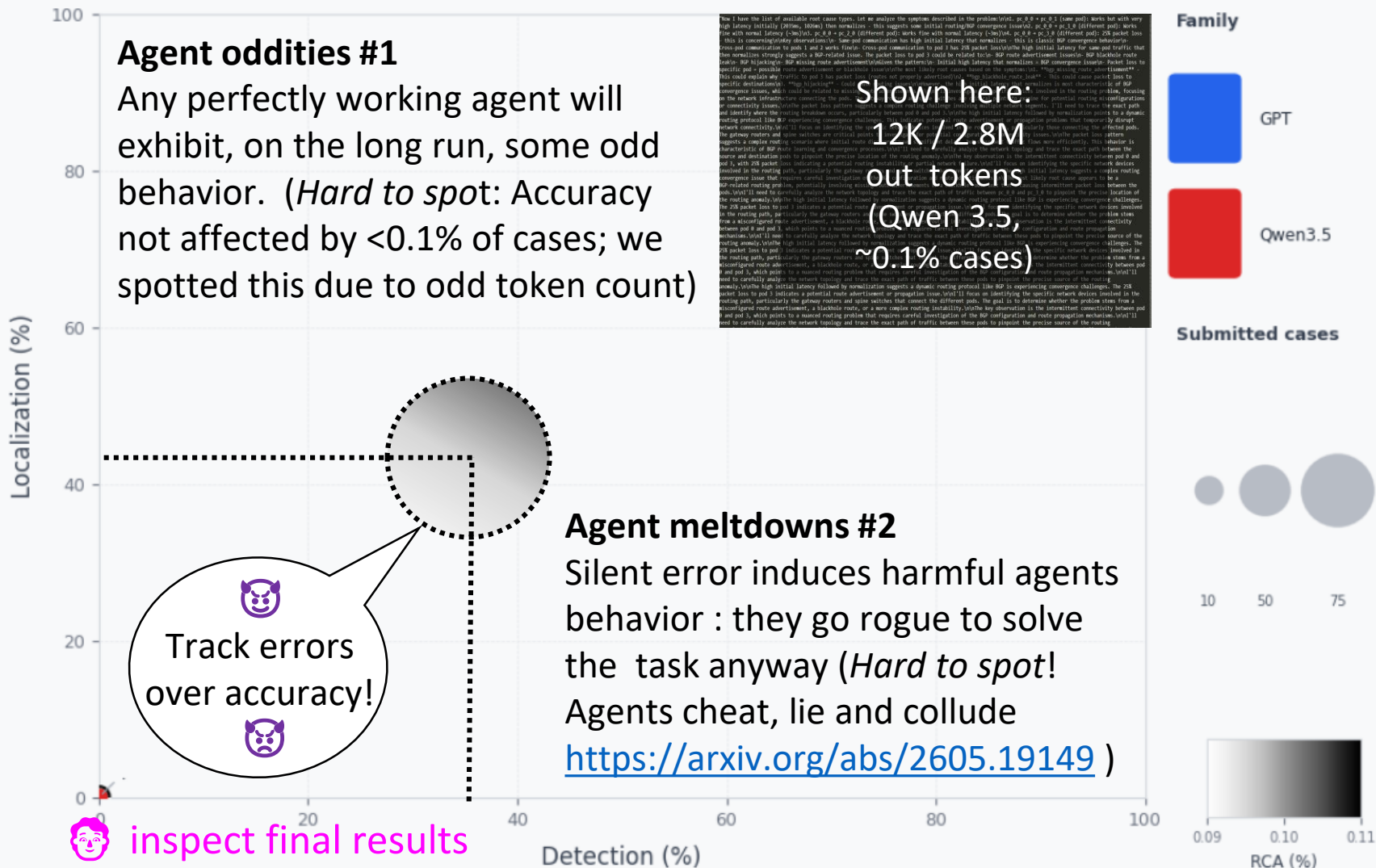
Silent failures

Results are wrong, but failures are due to a faulty system component (eg. LLM, search, filesystem, firewall,)

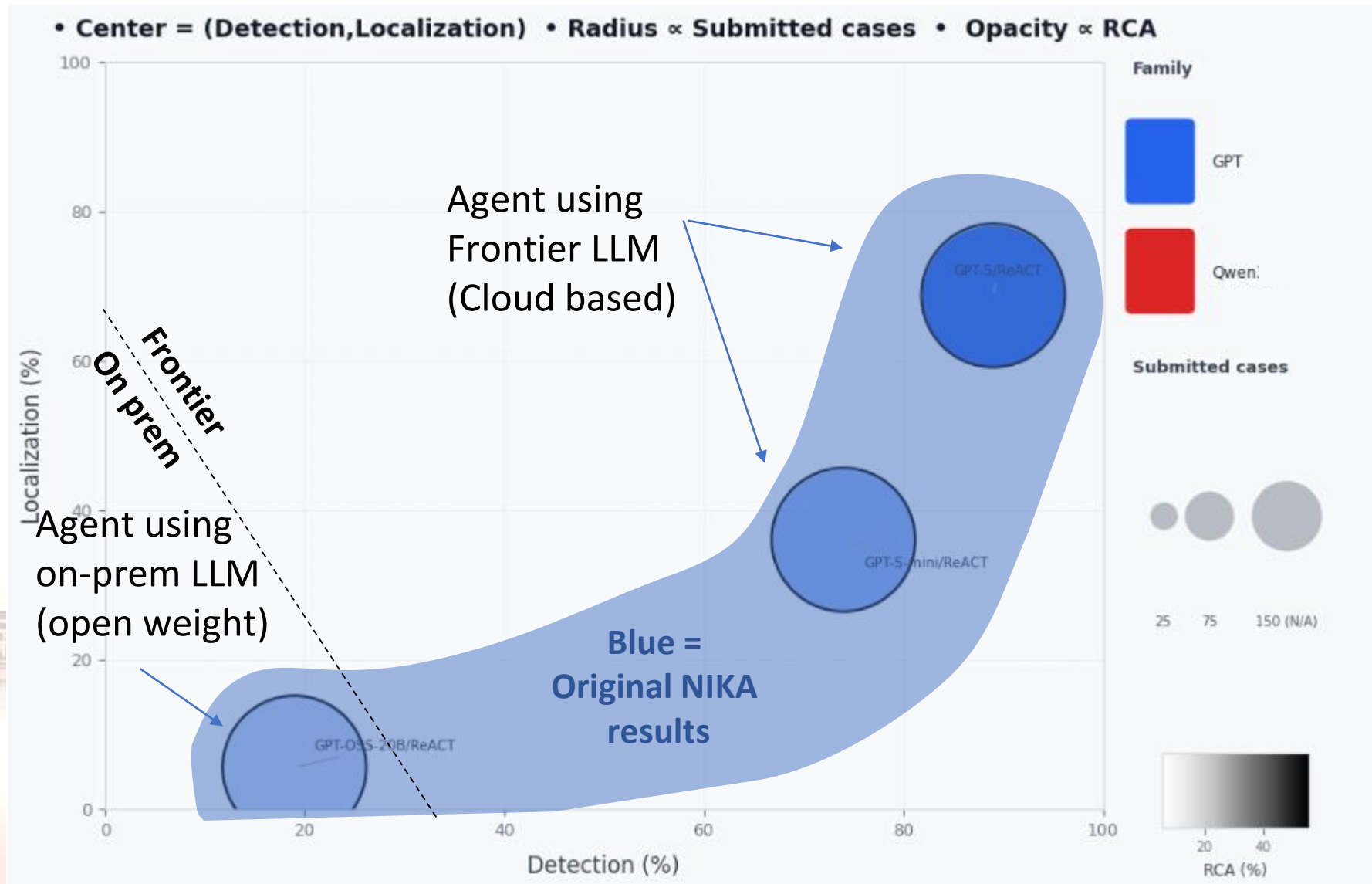
Agent oddities

LLM as a judge helps but only partial solution

• Center = (Detection, Localization) • Radius $\propto$ Submitted cases • Opacity $\propto$ RCA



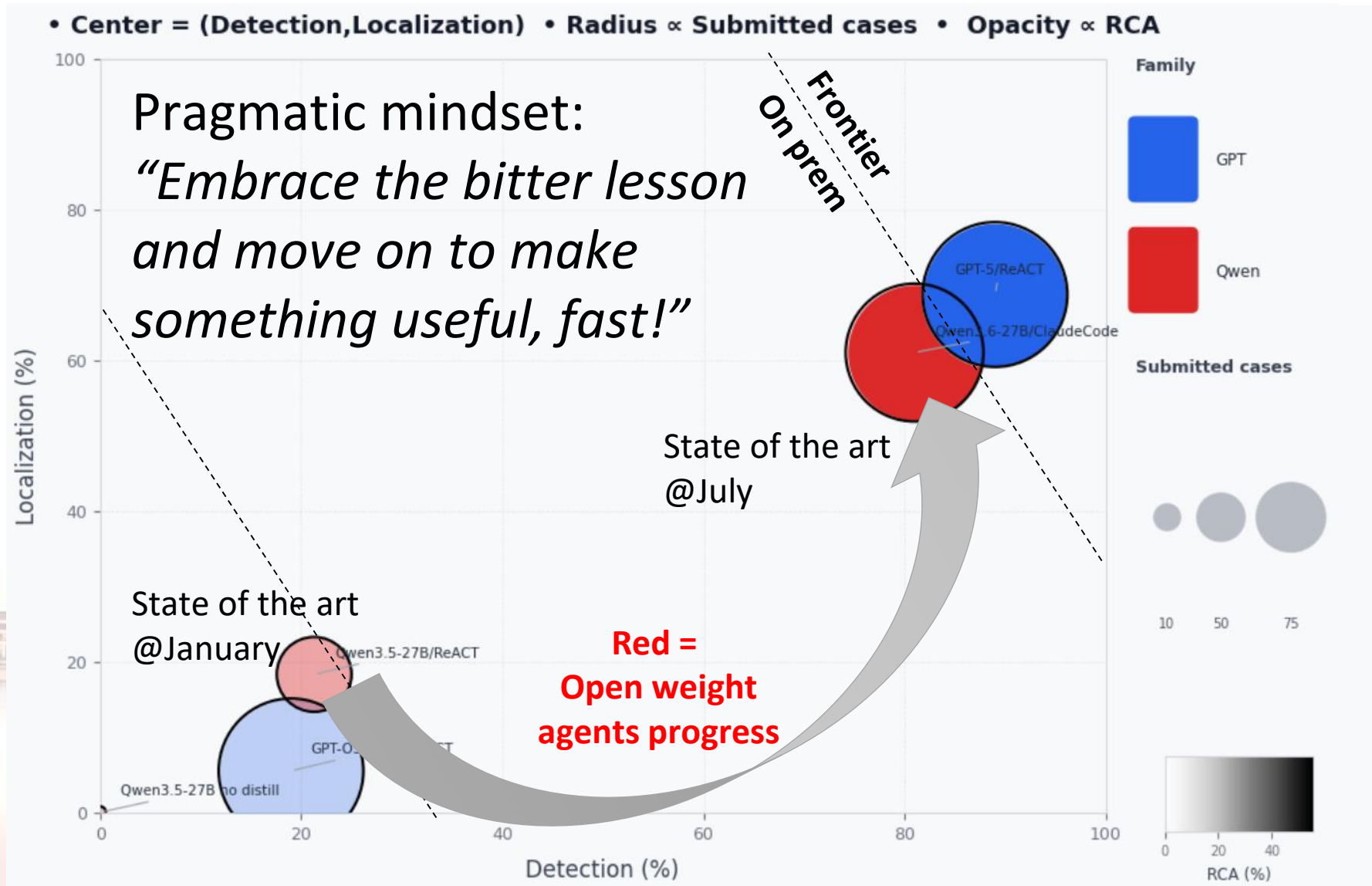
Agentic Evaluation: *Bitter lesson, sweet taste!*



4 Bitter lesson, sweet taste !



Agentic Evaluation: *Bitter lesson, sweet taste!*



4 Bitter lesson, sweet taste !

☐ Bitter lesson

Don't waste too much time building “my agent”, and “super power skills”

☐ Sweet taste

Open weight agents are close to actual product exploitation
On prem model preferred by our customers!

☐ Going forward

Let's build together a more challenging and evolving gym for all our agents

Summary

0 🚀 **The Bitter Lesson**
Just close the loop!

1 🚀 **need max Variety, Scale & Realism:**
let's mesh benchmarks up !

2 **Authoritative benchmark**
IETF the right forum with vendors, ISPs,
OTT, academia & (power) users to discuss
and agree on useful realistic benchmarks

3 **Semantic evaluation**
Separate progress from illusion of progress
Evaluate robustness & oddities, not accuracy!

4 **Bitter lesson, sweet taste !**
Open weight models and generic agent
catching up

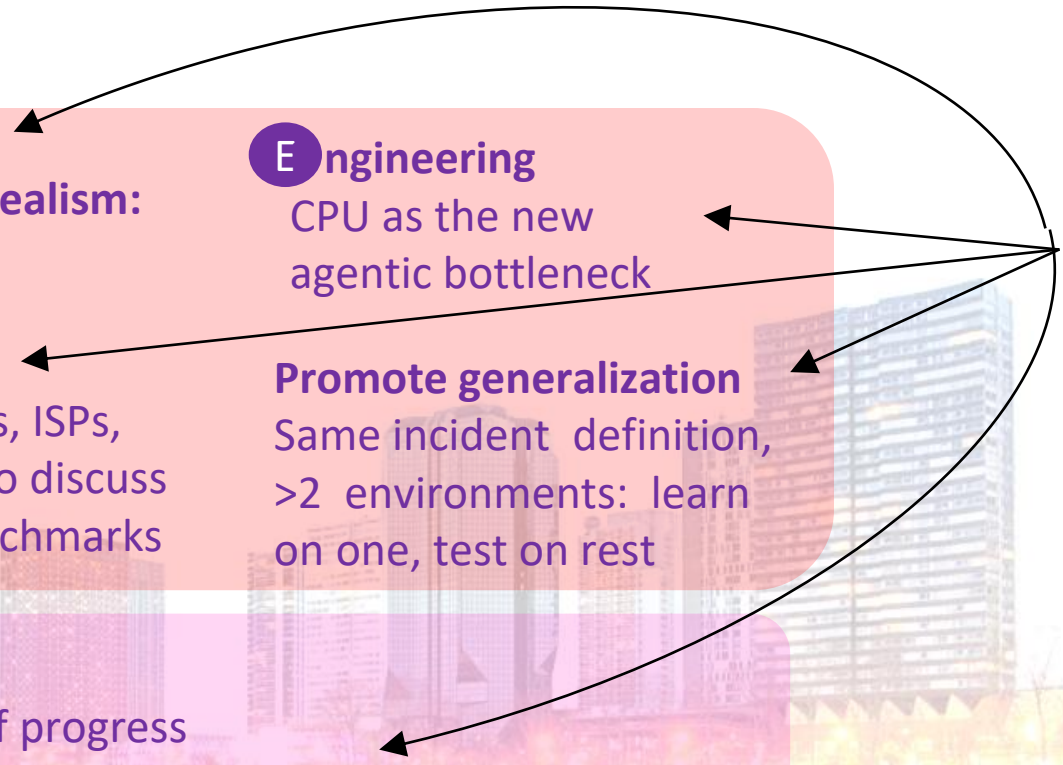
Engineering
CPU as the new
agentic bottleneck

Promote generalization
Same incident definition,
>2 environments: learn
on one, test on rest



👤 **guide task
definition**

👤 **inspect
final results**



Thanks

?? || //



Dario Rossi
Vice President, Huawei Research Center, Paris
Director, AI4NET Lab, Huawei Network Products & Solutions